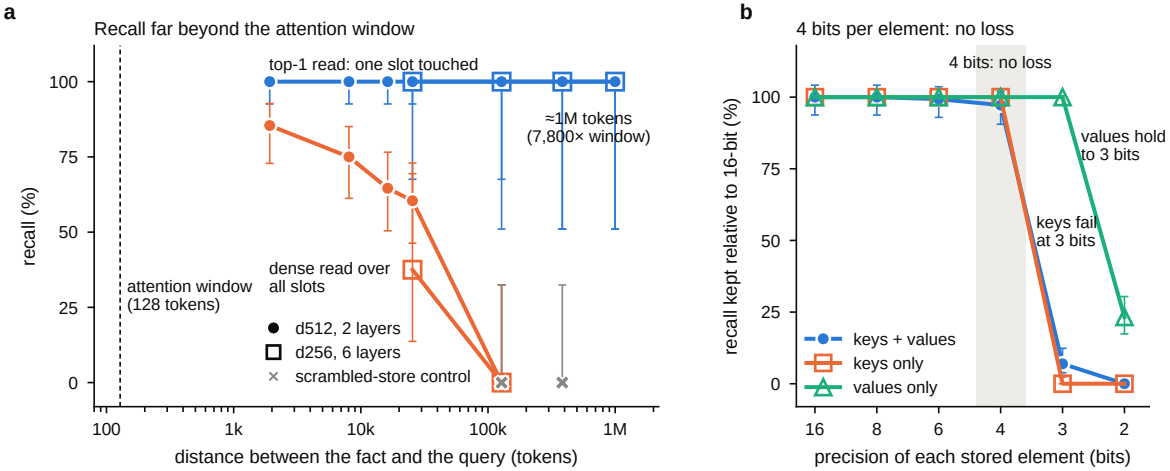
a
Recall far beyond the attention window
top-1 read: one slot touched
≈1M tokens
(7,800× window)
dense read over
all slots
attention window
(128 tokens)
d512, 2 layers
d256, 6 layers
scrambled-store control
recall (%)
100
75
50
25
0
100
1k
10k
100k
1M
distance between the fact and the query (tokens)
b
4 bits per element: no loss
4 bits: no loss
values hold
to 3 bits
keys fail
at 3 bits
keys + values
keys only
values only
recall kept relative to 16-bit (%)
100
75
50
25
0
16
8
6
4
3
2
precision of each stored element (bits)